Higher National Diploma in Engineering (Electrical)
4th Year, First Semester Examination – 2018
EE 3215 – Broadband Communication

Instructions for Candidates:	No. of questions	: 06
Answer any **FIVE (05)** questions	No. of pages	: 03
All Questions carry equal Marks	Time	: Three (03) hours

01.

i. "For residential broadband deployment in the country, DSL is the most cost-effective choice". Explain. **(05 Marks)**

ii. Identify the difference between the baseband and broadband communication methods used to transmit information over a network media. **(05 Marks)**

iii. Select the suitable and cost effective DSL technology (from given types) to fulfill each below requirement and give reasons for your selection.

1. ADSL Lite 2.SDSL 3. ADSL 4.VDSL 5.IDSL

a) Should be able to use for Internet and Web access, mostly for downloading films and web browsing, video on demand, remote LAN access; needs simultaneous POTS support, distance from the DSLAM is about 5km.

b) Should be able to use for Internet and Web access, at lower data rates: no need to watch videos or play games on line, do not want to install a splitter at customer premises; needs simultaneous POTS support, distance from the DSLAM is about 5km

c) Should be able to use for E1 service between web server and phone company or within a company; bidirectional-video, WAN, LAN, server access, but only one line of twisted-pair is available, no analog voice capabilities required hence does not need a splitter, distance from the DSLAM is about 3km

d) Should be able to use for voice & data as well as HDTV. Fiber to the Neighborhood connection is used. Distance from the DSLAM is less than 2km. Only one phone line to be used.

e) Should be able to support symmetrical rates up to 144kbps using existing telephone lines and ISDN terminal (ISDN modem), always ON data connection, but no standard telephone service. **(05 Marks)**

iv. Compare DSL technology with older data communication technologies like Dial-up modem and ISDN connections. **(05 Marks)**

[Total Marks 20]

02.

i. Explain how DSL technology achieves higher data rates over normal Copper wire pair. **(05 Marks)**

ii. Draw the end to end broadband connectivity diagram for a DSL connection and explain the function of each component involved in this connection. **(05 Marks)**

iii. Explain the two modulation techniques used in DSL and list the advantages of new technique over the old technique. **(05 Marks)**

iv. Explain Quadrature Amplitude Modulation (QAM) and draw constellation diagrams for 4-QAM and 16-QAM. **(05 Marks)**

[Total Marks 20]

03.

i. Suppose you have bought a new ADSL connection from your broadband service provider to upgrade your POTS connection to a Voice plus Data connection through existing cupper subscriber loop. You have been given an ADSL router with power supply, Splitter, two RJ11 cables, one RJ45 cable for you to install the ADSL router. Explain with a clear connectivity diagram, how these are interconnected so that you get the broadband connection to the computer in your room where you already have an extension telephone line with an ordinary telephone receiver (explain only the hardware connectivity, no need to explain the router configuration. Read the part ii of this question below, before you answer part i) **(06 Marks)**

ii. Once you have made the connection correctly in part (i) above and done the necessary software configuration correctly, you got the broadband connection working successfully and you could access the internet. But you have noticed that the noise level on the main voice connection (the main voice connection is at the sitting room, your room has a parallel connection to it) has gone up and the data rate of ADSL connection is slower than what you have been promised by the service provider. So you made a complaint and the service provider, after testing the line, mentioned that the quality of the line is poor. And also they confirmed that there is no problem in the external loop up to your house with regard to the noise and distance. Explain the possible causes for this situation and suggest how you can overcome this problem and get the best possible speed of the data connection without disturbing the voice connection. **(06 Marks)**

iii. CPE (ADSL MODEM) can be configured in two different modes; Transparent Bridge mode and Router Mode. Briefly explain these two modes using two connectivity diagrams. **(05 Marks)**

iv. Give a brief description about Asynchronous Transfer Mode (ATM). **(03 Marks)**

[Total Marks 20]

04.

i. What is XTALK in UTP cables used in DSL? **(05 Marks)**

ii. How can the effect of XTALK be minimized in DSL communication? **(05 Marks)**

iii. What is a loading coil and how does it affect DSL signals? Explain. **(05 Marks)**

iv. Give three constraints that prevents the effective transmission of DSL signals on Copper wires and explain how the negative impact of each constraint can be minimized?

(05 Marks)

[Total Marks 20]

05.

Broadband Service Providers differentiate Asymmetric version of DSL (ADSL) from Symmetric version (SDSL) in their billing criteria towards customers. i.e They charge at lower rates for ADSL home users whereas corporate and commercial users who use SDSL are charged at higher rates. Explain commercial and technical reasons behind this differentiation in the view point of the Service Provider. **(05 Marks)**

i. What is Cable Modem technology in wired broadband access? Some people say, Cable Modem is faster than DSL. Do you agree it? Explain with reasons. **(05 Marks)**

ii. ADSL still stands popular among alternate high speed internet access technologies using wired lines. But when it comes to wireless broadband technologies, ADSL has to face strong competition. Discuss. **(05 Marks)**

iii. BPL should be a better option for broadband access than DSL. But in practical implementation, BPL has an issue using commercial power line as a broadband access medium. Discuss.

(05 Marks)

[Total Marks 20]

06.

i. Explain the two modes of WiMAX connectivity with a network diagram. **(05 Marks)**

ii. What are the possible situations in which data communication over ADSL works even without dial tone on the telephone receiver? **(05 Marks)**

iii. What will you observe on data connection and voice connection, if the splitter is not used in ADSL enabled telephone line **(05 Marks)**

iv. Explain how Wi-Fi differs from WiMAX. **(05 Marks)**

[Total Marks 20]